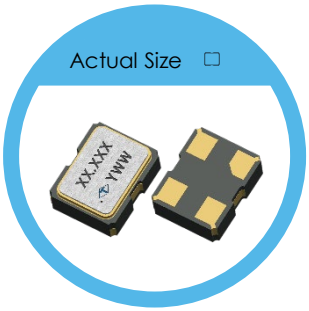
TAITIEN
Our Performance, Your Reputation

OZ Type
2.0 x 1.6 mm SMD Crystal Oscillator



RoHS Compliant

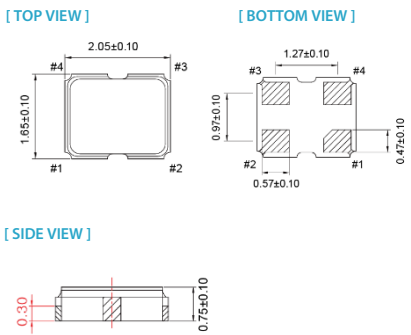
FEATURES

- Conforms to AEC-Q200
- Typical 2.05 x 1.65 x 0.75 mm ceramic SMD package
- Tight symmetry (45 to 55%) available
- Operation voltage: 1.8V, 2.5V, 3.3V
- Tri-state enable/disable

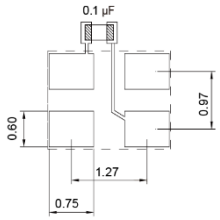
TYPICAL APPLICATION

- WLAN/WiMax
- Mobile Phone
- DSC,Set-top Box,HDTV

DIMENSION (mm)



SOLDER PAD LAYOUT (mm)



To ensure optimal oscillator performance, place a by-pass capacitor of 0.1µF as close to the part as possible between V_{DD} and GND pads.

PIN FUNCTION (mm)

PIN#	FUNCTION
1	Tri-State
2	GND
3	Output
4	V_{DD}

ELECTRICAL SPECIFICATION

Parameter		3.3V		2.5V		1.8V		Unit	Test Condition
		Min.	Max.	Min.	Max.	Min.	Max.		
Supply Voltage Variation (V_{DD})		V_{DD} – 10%	V_{DD} + 10%	V_{DD} – 10%	V_{DD} + 10%	V_{DD} – 10%	V_{DD} + 10%	V	
Frequency Range		1.5	50	1.5	50	1.5	50	MHz	
Standard Frequency		24, 26						MHz	Standard frequencies are frequencies which the crystal has been designed and does not imply a stock position.
Supply Current			15		10		7	mA	
Duty Cycle		45	55	45	55	45	55	%	
Output Level (CMOS)	Output High (Logic "1")	2.97		2.25		1.62		V	
	Output Low (Logic "0")		0.33		0.25		0.18	V	
Transition Time: Rise/Fall Time	1.5 MHz ≦ Fo < 20 MHz		4		4		5	nSec	Transition times are measured between 10% and 90% of V_{DD}, with an output load of 15 pF.
	20 MHz ≦ Fo < 50 MHz		3		3		4	nSec	
Start Time			2		2		2	mSec	
Tri-State (Input to Pin 1)	Enable (High Voltage or Floating)	2.31		1.75		1.26		V	
	Disable (Low Voltage or GND)		0.99		0.75		0.54	V	
Period Jitter (pk-pk)			40		40		40	pSec	
RMS Phase Jitter (Integrated 12 kHz~20 MHz)			1		1		1	pSec	
Standby Current			10		10		10	µA	
Aging (@25°C 1st year)			±3		±3		±3	ppm	
Storage Temp. Range		-55	+125	-55	+125	-55	+125	°C	

FREQ. STABILITY vs. TEMP. RANGE

ppm / Temp.(°C)	±20	±25	±50
-10 ~ +60	○	○	○
-20 ~ +70	△	○	○
-40 ~ +85	×	△	○
-40 ~ +125	×	×	△

○: Available △:Conditional ×: Not Available
Inclusive of calibration @ 25°C ,operating temperature rrange,input Voltage variation,load variation,aging (1st year),shock,and vibration

Note: not all combination of options are available. Other specifications may be available upon request.
Specifications subject to change without notice.

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